

S&P 500 Profitability: Identifying Characteristics of High-Performing Sectors (2013-2018)

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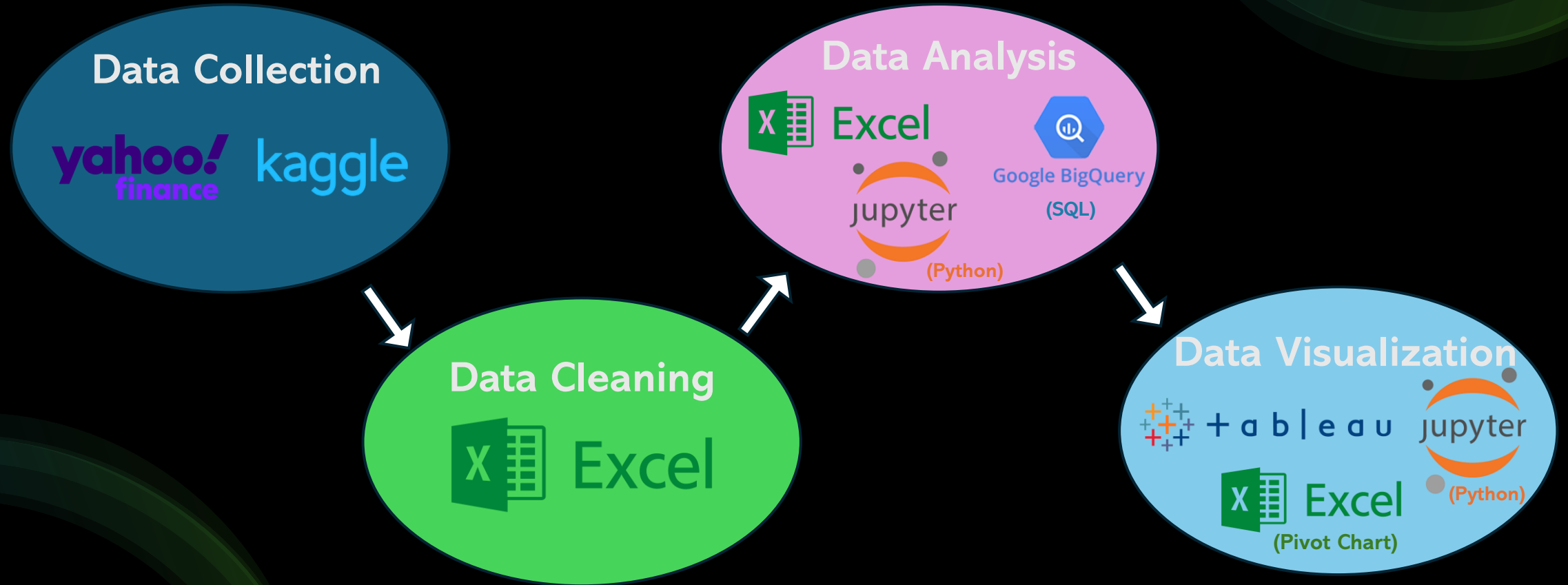


Objective for Data Analysis

Which sectors and stock characteristics (KPIs) generated the strongest profitability from the S&P 500 from 2013-2018?



Data Analysis Process



Data Overview

Metric	Value
Stocks	505
Daily Observations	619,040
Time Period	2/8/2013-2/7/2018
Sources	Kaggle (all_stocks_5yr.csv By JANIO MARTINEZ BACHMANN) Yahoo Finance S&P500 Historical Data
Columns	Date, Open, High, Low, Close, Volume, Name

Sample of Dataset (Kaggle)

	A	B	C	D	E	F	G
1	date	open	high	low	close	volume	Name
2	2/8/2013	15.07	15.12	14.63	14.75	8407500	AAL
3	2/11/2013	14.89	15.01	14.26	14.46	8882000	AAL
4	2/12/2013	14.45	14.51	14.1	14.27	8126000	AAL
5	2/13/2013	14.3	14.94	14.25	14.66	10259500	AAL
6	2/14/2013	14.94	14.96	13.16	13.99	31879900	AAL
7	2/15/2013	13.93	14.61	13.93	14.5	15628000	AAL
8	2/19/2013	14.33	14.56	14.08	14.26	11354400	AAL
9	2/20/2013	14.17	14.26	13.15	13.33	14725200	AAL
10	2/21/2013	13.62	13.95	12.9	13.37	11922100	AAL
11	2/22/2013	13.57	13.6	13.21	13.57	6071400	AAL

Sample of Dataset (Yahoo Finance)

Date	Open	High	Low	Close ⓘ	Adj Close ⓘ	Volume
Feb 21, 2013	1,511.95	1,511.95	1,497.29	1,502.42	1,502.42	4,274,600,000
Feb 20, 2013	1,530.94	1,530.94	1,511.41	1,511.95	1,511.95	4,240,570,000
Feb 19, 2013	1,519.79	1,530.94	1,519.79	1,530.94	1,530.94	3,748,910,000
Feb 15, 2013	1,521.38	1,524.24	1,514.14	1,519.79	1,519.79	3,838,510,000
Feb 14, 2013	1,520.33	1,523.14	1,514.02	1,521.38	1,521.38	3,759,740,000
Feb 13, 2013	1,519.43	1,524.69	1,515.93	1,520.33	1,520.33	3,385,880,000
Feb 12, 2013	1,517.01	1,522.29	1,515.61	1,519.43	1,519.43	3,414,370,000
Feb 11, 2013	1,517.93	1,518.31	1,513.61	1,517.01	1,517.01	2,684,100,000
Feb 8, 2013	1,509.39	1,518.31	1,509.39	1,517.93	1,517.93	2,986,150,000

Top Individual Performers

Most Profitable Stocks

*Found via **BigQuery(SQL)**

Ticker	Name	Sector	Profit (%)
NVDA	NVIDIA Corp.	Information Technology	+1750%
NFLX	Netflix Inc.	Communication Services	+923%
ALGN	Align Technology Inc.	Healthcare	+616%
EA	Electronic Arts Inc.	Consumer Discretionary	+608%
STZ	Constellation Brands Inc.	Information Technology	+572%

Bottom Individual Performers

Least Profitable Stocks

****Company has merged since 2018**

*Found via **BigQuery(SQL)**

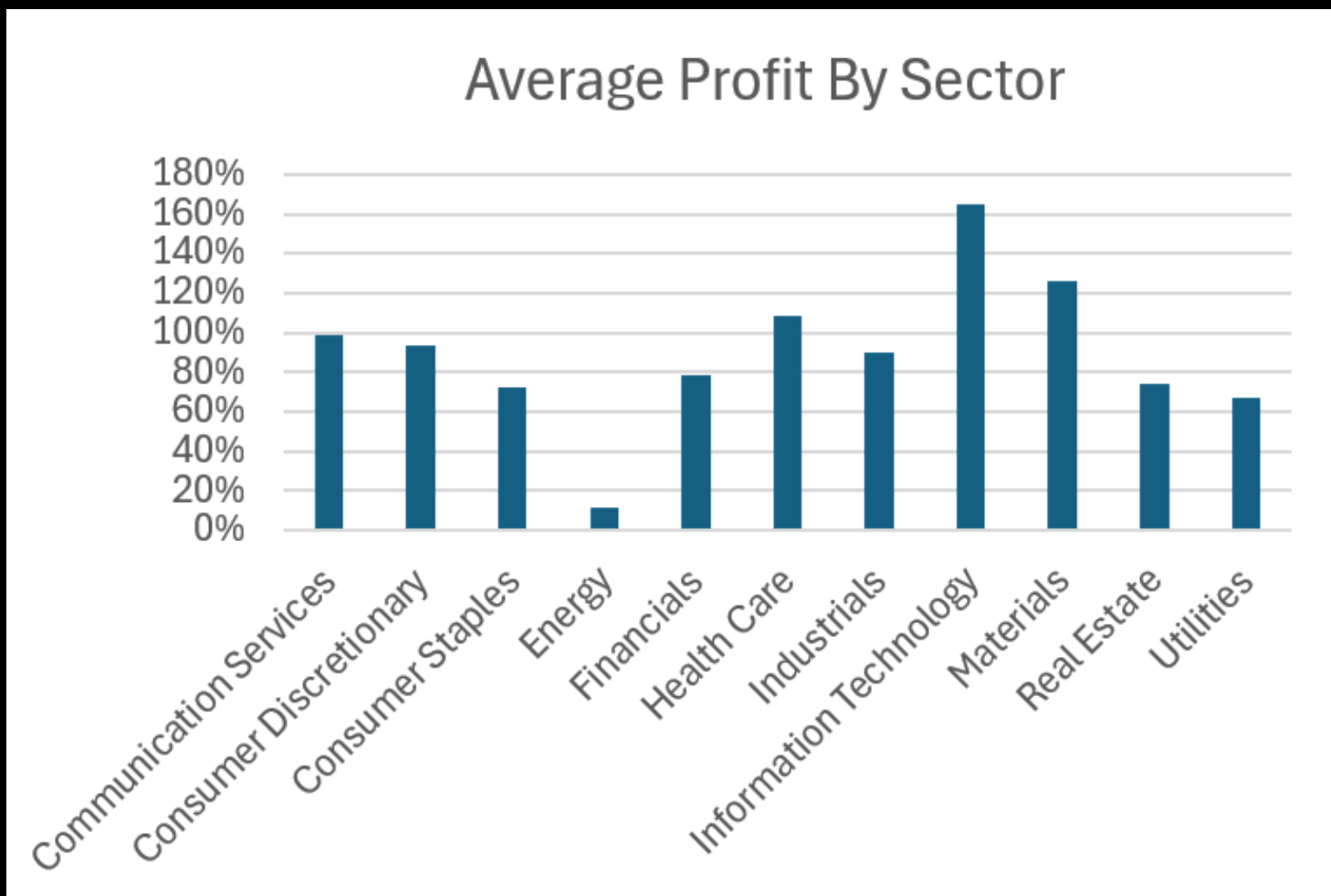
Ticker	Name	Sector	Profit (%)
CHK	Chesapeake Energy **	Energy	-86%
RRC	Range Resources Corp.	Energy	-82%
UA	Under Armour Inc.	Consumer Discretionary	-72%
DISCA	Discovery Inc. **	Communication Services	-68%
DISCK	Discovery Inc. **	Communication Services	-66%

Profit By Sector

- **Top 3 Sectors:**
 - 1) Information Technology (+164.9%)
 - 2) Materials (+125.6%)
 - 3) Healthcare (+108.2%)
- **Bottom 3 Sectors:**
 - 1) Energy (+12%)
 - 2) Utilities (+66.6%)
 - 3) Consumer Staples (+72%)

*Found via **BigQuery(SQL)**

*Visualization: **Excel Pivot Chart**

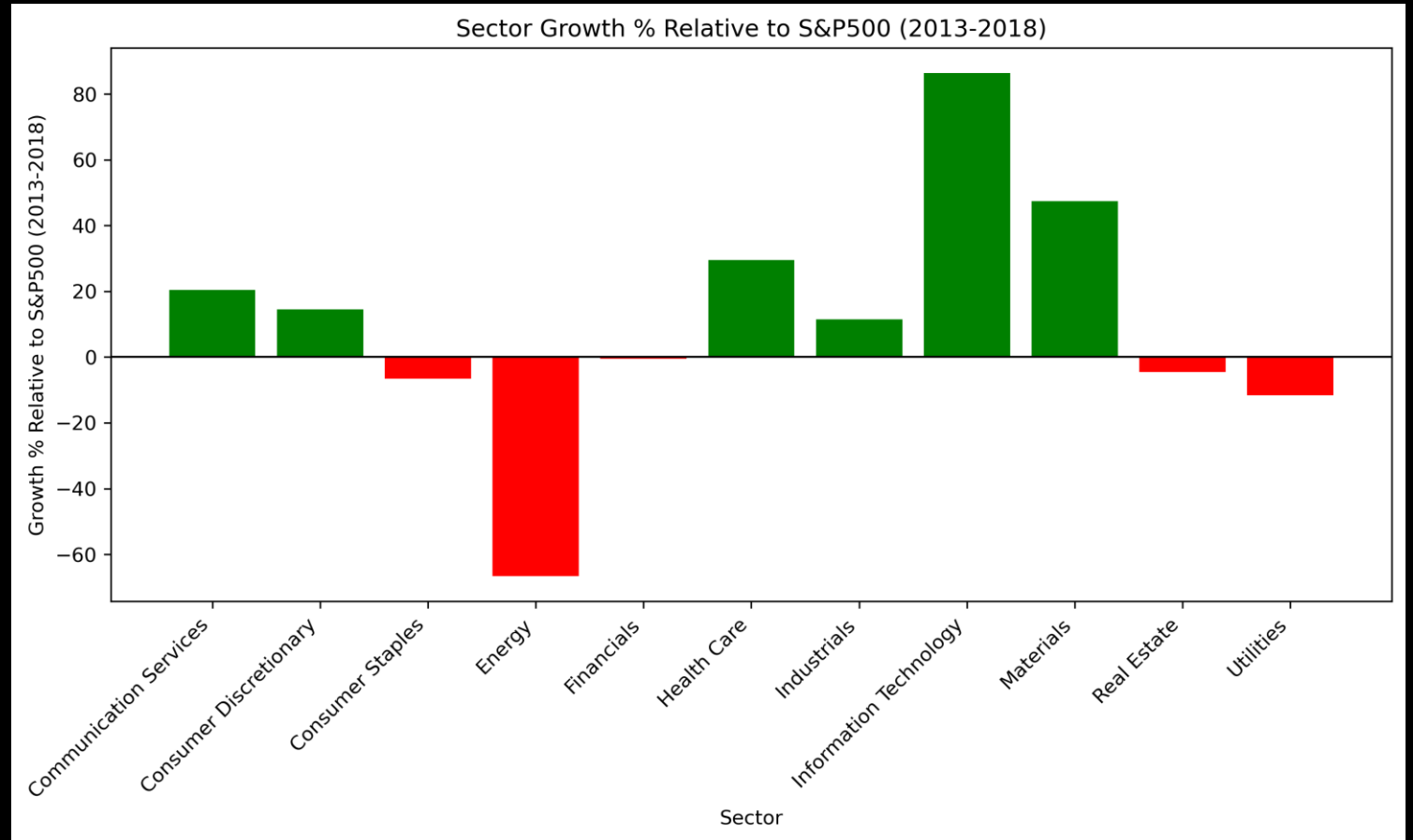


Profit By Sector Relative to S&P500

- **S&P500 Overall Profit: 78.6%**
- **Top 3 Sectors:**
 - 1) Information Technology **(+86.4%)**
 - 2) Materials **(+47.4%)**
 - 3) Healthcare **(+29.4%)**
- **Bottom 3 Sectors:**
 - 1) Energy **(-66.6%)**
 - 2) Utilities **(-11.6%)**
 - 3) Consumer Staples **(-6.6%)**

*Found via **Python (Pandas Library)**

*Visualization: **Matplotlib via Python**



Does Trading Volume Influence Stock Profitability?

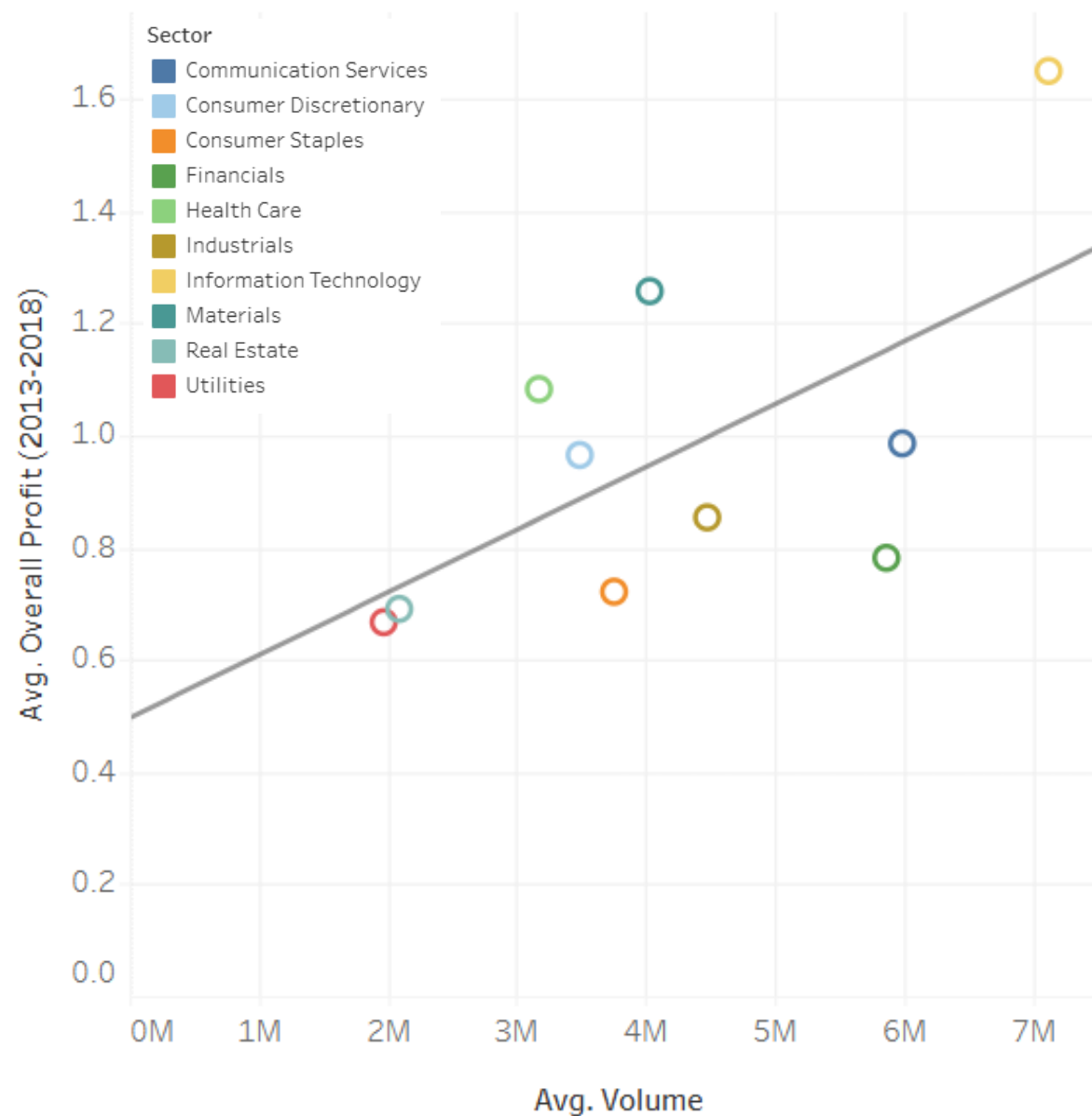
R-Squared: 0.38

P-value: 0.05

Statistically Significant

*Visualization: Tableau

Avg. Profit vs. Avg. Volume (5-year) By Sector



Does Higher Risk Lead to Higher Profit?

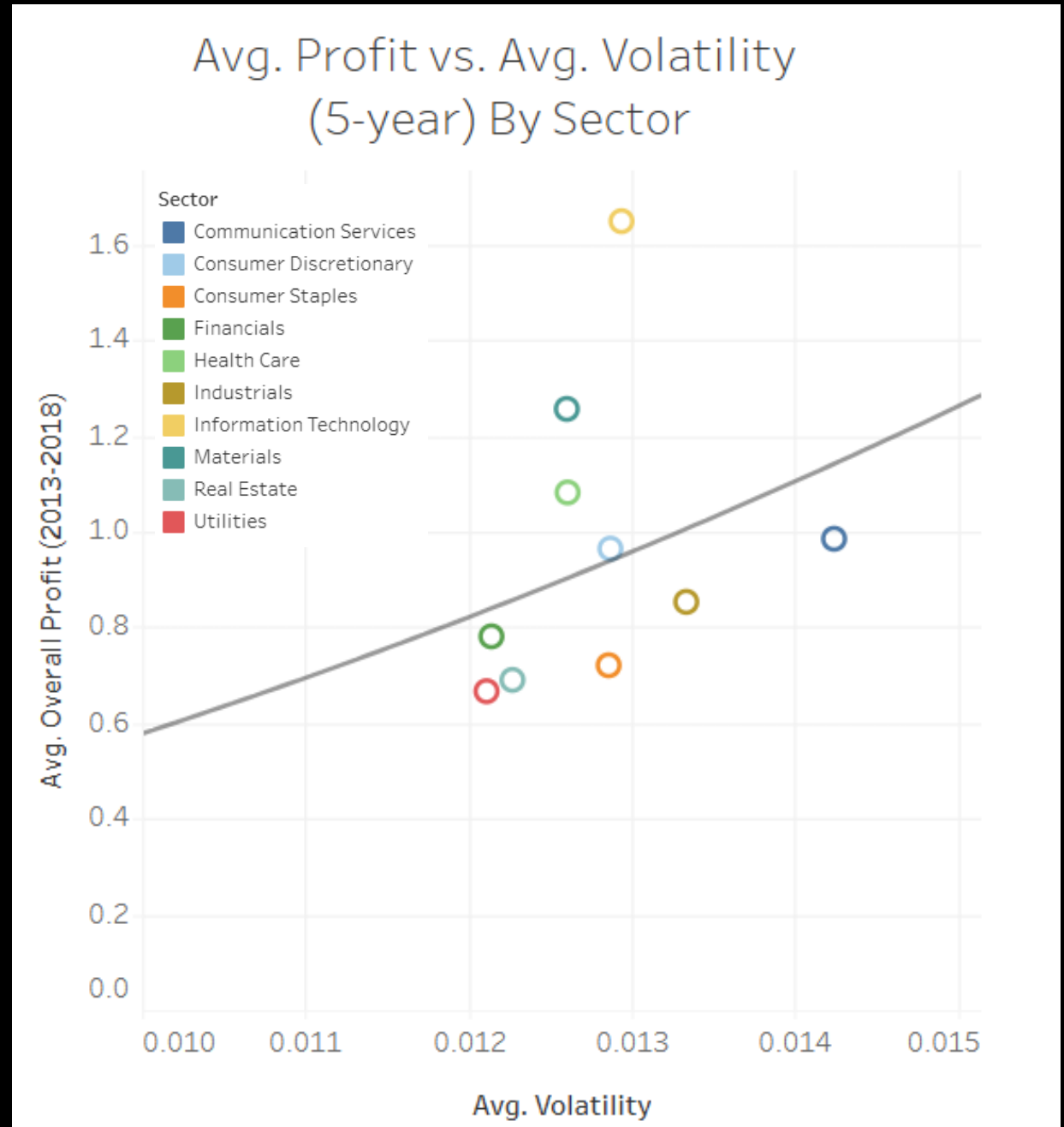
R-Squared: 0.10

P-value: 0.35

Not Statistically Significant

Volatility = Standard
Deviation of
Average Daily
Returns

*Visualization: Tableau



Does Win Rate (%) Influence Profit?

R-Squared: 0.48

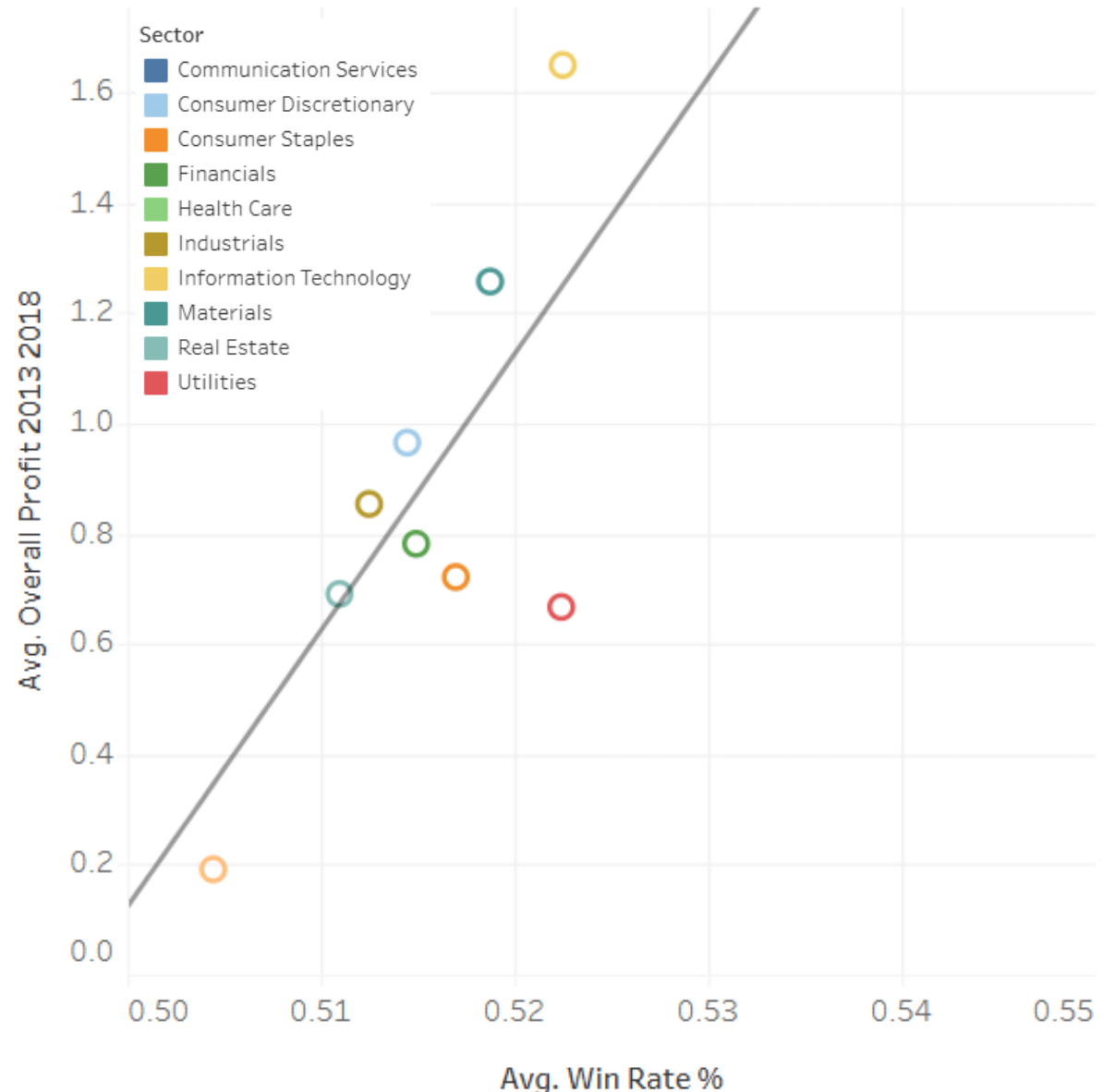
P-value: 0.02

Statistically Significant

$$\text{Win Rate (\%)} = \frac{\text{Positive Trading Days}}{\text{Total Trading Days}}$$

*Visualization: Tableau

Avg. Profit vs. Avg. Win Rate (%) By Sector



Analysis Summary

After my data analysis of the S&P 500 from 2013-2018, I can conclude from this data that:

- The **most profitable sector** was Information Technology (+165%)
**Outperformed the S&P500 by +86.4%*
- The **most profitable position** was NVIDIA Corp. (+1750%)
- The **least profitable sector** was Energy (+12%)
**Underperformed the S&P500 by -66.6%*
- The **least profitable position** was Chesapeake Energy (-86%)
- Volume and Profit have a **moderately positive linear correlation**
- Volatility and Profit **do not have any significant correlation**
- Win Rate (%) and Profit have a **moderately positive linear correlation**

Investor Recommendations

- Investors should consider **historical sector performance**, since some sectors performed substantially better than others
- **Trading volume** may serve as an indicator of future profit for investors
- **Win Rate (%)** may serve as an indicator of future profit for investors

For Future Research:

- More **recent** market data should be incorporated to determine whether these historical relationships persist under current markets.

Appendices (SQL Scripts)

Scripts for Most/Least Profitable Stocks on Avg.

```
SELECT
    Name,
    Sector,
    CAST((Overall_Profit_2013_2018)*
100 AS INT64) AS
Overall_Profit_Percentage

FROM `mystic-benefit-497822-
t2.stock_data.stock_data_cleaned`

ORDER BY
    Overall_Profit_2013_2018 DESC

LIMIT 5
```

```
SELECT
    Name,
    Sector,
    CAST((Overall_Profit_2013_2018)*
100 AS INT64) AS
Overall_Profit_Percentage

FROM `mystic-benefit-497822-
t2.stock_data.stock_data_cleaned`

ORDER BY
    Overall_Profit_2013_2018 ASC

LIMIT 5
```

Appendices (SQL Scripts)

Scripts for Most/Least Profitable Sectors on Avg.

```
SELECT
    Sector,
    (100 * AVG(Overall_Profit_2013_2018)) AS
Overall_Profit_Percentage
FROM `mystic-benefit-497822`
t2.stock_data.stock_data_cleaned`
GROUP BY
    Sector
ORDER BY
    Overall_Profit_Percentage DESC
```

```
SELECT
    Sector,
    (100 * AVG(Overall_Profit_2013_2018)) AS
Overall_Profit_Percentage
FROM `mystic-benefit-497822`
t2.stock_data.stock_data_cleaned`
GROUP BY
    Sector
ORDER BY
    Overall_Profit_Percentage ASC
```

Appendices (Python Scripts)

Scripts for Most/Least Profitable Sectors relative to the S&P500

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import yfinance as yf
sp500 = yf.download('^GSPC', start = '2013-2-8', end = '2018-2-7')

firstDayOpen = (sp500['Open'].iloc[0]).item()
lastDayClose = sp500['Close'].iloc[-1].item()
overallProfitPercent = round(((lastDayClose-firstDayOpen)/firstDayOpen)*100,2)
overallProfitPercent

sectorPerformance = pd.read_csv('Stock_Profitability_Data_Raw.csv')
sectorPerformance = sectorPerformance.head(11)
sectorPerformance
```


Appendices (Python Scripts)

Scripts for Most/Least Profitable Sectors relative to the S&P500

```
sectorPerformance['Growth_Relative_to_S&P'] = sectorPerformance['Average of  
Overall_Profit_2013_2018'] - (overallProfitPercent)  
sectorPerformance  
  
growth = sectorPerformance["Growth_Relative_to_S&P"]  
# Green for positive, red for negative  
colors = ["green" if value >= 0 else "red" for value in growth]  
plt.figure(figsize=(10,6))  
plt.bar(    sectorPerformance["Sector"],growth,color=colors)  
plt.title("Sector Growth % Relative to S&P500 (2013-2018)")  
plt.xlabel("Sector")  
plt.ylabel("Growth % Relative to S&P500 (2013-2018)")  
plt.xticks(rotation=45, ha="right")plt.axhline(0, color="black", linewidth=1)
```

Appendices (Python Scripts)

Scripts for Most/Least Profitable Sectors relative to the S&P500

```
plt.tight_layout()  
plt.savefig("sector_growth.png", dpi=300, bbox_inches="tight")  
plt.show()
```